tracking systems to acquire and navigate to targets with greater precision and control.

- Varying the size of nanometals, such as nanoaluminum, will give us control over the explosion of munition and thereby minimise collateral damage. Incorporating nanometals into bombs and propellants increases the speed of released energy with fewer raw materials consumed.
- Chemical sensors based on nanotechnology will be incredibly sensitive – capable, in fact, of pinpointing a single molecule out of billions. These sensors will be cheap and disposable, forewarning us of airport-security breaches or anthrax-laced letters. These sensors will eventually take to the air on military unmanned aerial vehicles (UAVs), not only sensing chemicals but also providing incredible photo resolutions. These photos, condensed and on an energy-efficient, high resolution, wristwatch-sized display, will find their way to the soldier, providing incredible real-time situational awareness at the place needed most: the front lines.

Computing


Levitation is the way to go

- Today, computer chips are made using lithography – literally, "stone writing". If the computer hardware revolution is to continue at its current pace, in a decade or so we'll have to move beyond lithography to some new post-lithographic manufacturing technology. Ultimately, each logic element will be made from just a few atoms. Designs for computer gates with less than 1,000 atoms have already been proposed – but each atom in such a small device has to be in exactly the right place. To economically build and interconnect trillions upon trillions of such small and precise devices in a complex three-dimensional pattern we'll need a manufacturing technology well beyond today's lithography: we'll need nanotechnology.
- We'll be able to build mass storage devices that can store more than a hundred billion billion bytes in a volume the size of a sugar cube; RAM